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Dr. M.G.R.
EDUCATIONAL AND RESEARCH INSTITUTE
DEEMED TO BE UNIVERSITY

University with Graded Autonomy Status

(An ISO 21001 : 2018 Certified Institution)

Periyar E.V.R. High Road, Maduravoyal, Chennai-95, Tamilnadu, India.



**RetinAI: AN AI-POWERED CNN-BASED SYSTEM FOR
MULTI-CLASS EYE DISEASE DETECTION AND
EXPLAINABLE DIAGNOSTIC REPORTING**

MINI PROJECT REPORT

submitted in partial fulfillment of the requirements

for the award of the degree in

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING(ARTIFICIAL INTELLIGENCE)

by

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**DEPARTMENT OF
ARTIFICIAL INTELLIGENCE**

AUGUST 2026



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DEPARTMENT OF ARTIFICIAL INTELLIGENCE

BONAFIDE CERTIFICATE

This is to certify that this Mini Project Report titled “**RetinAI: An AI-Powered CNN-Based System for Multi-Class Eye Disease Detection and Explainable Diagnostic Reporting**” is the bonafide work of Mr. **J SRI HARI (231211101346)**, Mr. **K S V ABHINASH (231211101351)**, and Mr. **SIVA BHARATH (231211101289)**, who carried out the Mini Project under our supervision from February 2026 to July 2026.

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MAJOR DESIGN CONSTRAINTS AND DESIGN STANDARDS

Field	Details
Student Group	J SRI HARI 231211101346 K S V ABHINASH 231211101351 SIVA BHARATH 231211101289
Project Title	VisionCheck: An AI-Powered CNN-Based System for Multi-Class Eye Disease Detection and Explainable Diagnostic Reporting
Program Concentration Area	Medical Image Analysis, Deep Learning, Computer Vision, Web-Based Healthcare Applications
Constraints Examples	Variability in fundus/OCT image quality (illumination, resolution, noise); limited compute for real-time browser-based inference; class imbalance across seven diagnostic categories; need for model interpretability in a clinical setting; privacy of patient images and scan history
Economic	Yes
Environmental	Yes
Sustainability	Yes
Implementable	Yes
Ethical	Yes
Health and Safety	Yes
Social	Yes
Political	NA
Other	NA
Standards	1. IEC 62304 – Medical Device Software Life-Cycle Processes 2. ISO/IEC 25010 – Systems and Software Quality Requirements and Evaluation (SQuaRE) 3. IEEE 830 – Recommended Practice for Software Requirements Specifications 4. ISO/IEC 27001 – Information Security Management Systems 5. HIPAA / India's Digital Information Security in Healthcare Act (DISHA) — health data privacy guidelines

Prerequisite Courses for the Mini Project

1. Machine Learning
2. Deep Learning / Neural Networks
3. Digital Image Processing
4. Python Programming
5. Web Technologies

ABSTRACT

Eye diseases such as diabetic retinopathy, glaucoma, cataract, and age-related macular degeneration are among the leading causes of preventable blindness worldwide, and early detection is critical to limiting irreversible vision loss. In many regions, however, access to a qualified ophthalmologist for routine retinal screening remains limited, creating a strong case for automated, AI-assisted pre-screening tools. This project presents VisionCheck, a browser-based, deep-learning-driven system that classifies retinal, fundus, OCT, and iris images into seven diagnostic categories: diabetic retinopathy, cataract, glaucoma, age-related macular degeneration, conjunctivitis, retinal detachment, and normal (no pathological finding).

The system uses a convolutional neural network (CNN) built on transfer learning, with EfficientNetB0 and ResNet50 as candidate base architectures fine-tuned on a combination of the ODIR, EyePACS, MESSIDOR-2, Kaggle DR, STARE, and DRIVE datasets, and a lightweight MobileNetV2 model as a fallback for constrained, client-side inference. Input images undergo a preprocessing pipeline consisting of resizing to 224×224, Contrast Limited Adaptive Histogram Equalization (CLAHE), noise reduction, normalisation, and augmentation before being passed through the network. To make the model's predictions interpretable to clinicians, the system incorporates Gradient-weighted Class Activation Mapping (Grad-CAM), which highlights the retinal regions that most influenced a given diagnosis.

Beyond classification, VisionCheck generates a structured clinical insight report for each prediction — covering probable causes, risk factors, symptoms, evidence-informed treatment options, preventive precautions, and a specialist-referral recommendation — along with a confidence score and a probability distribution across all seven classes. An analytics dashboard summarises disease distribution, confidence trends, severity breakdown, and model performance metrics (accuracy, precision, recall, F1-score, specificity, and AUC-ROC) across a user's scan history. The prototype achieves an overall validation accuracy of approximately 79% using the EfficientNetB0/ResNet50 configuration.

The system is explicitly positioned as a preliminary screening aid rather than a diagnostic replacement, with a persistent medical disclaimer directing users to a qualified ophthalmologist for confirmation. All image processing in the deployed client-side demonstration is performed locally in the browser using TensorFlow.js, without transmitting patient images to an external server, in keeping with privacy-by-design principles relevant to healthcare data. This report documents the motivation, literature background, system design, implementation, and evaluation of the VisionCheck system, and outlines directions for extending it toward clinical-grade deployment.

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LIST OF SYMBOLS, ABBREVIATIONS AND NOMENCLATURE

AI	Artificial Intelligence
AMD	Age-related Macular Degeneration
API	Application Programming Interface
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
CLAHE	Contrast Limited Adaptive Histogram Equalization
CNN	Convolutional Neural Network
DR	Diabetic Retinopathy
Grad-CAM	Gradient-weighted Class Activation Mapping
HbA1c	Glycated Haemoglobin
IOP	Intraocular Pressure
OCT	Optical Coherence Tomography
ODIR	Ocular Disease Intelligent Recognition (dataset)
ReLU	Rectified Linear Unit
UML	Unified Modelling Language
VEGF	Vascular Endothelial Growth Factor

1. INTRODUCTION

1.1 BROAD AREA OF THE PROJECT

The project falls within the broad area of medical image analysis using deep learning, specifically the application of convolutional neural networks (CNNs) to ophthalmic screening. It combines elements of computer vision, transfer learning, explainable artificial intelligence (XAI), and web-based healthcare application design to build an end-to-end system that can classify eye images into clinically relevant categories and present the result in a form that is useful to both patients and clinicians.

1.2 PROBLEM STATEMENT

Vision-threatening eye diseases such as diabetic retinopathy, glaucoma, and age-related macular degeneration are frequently asymptomatic in their early, most treatable stages. Effective management therefore depends on regular screening, but comprehensive manual screening — reading retinal photographs one by one — does not scale to the size of at-risk populations, particularly in regions with a shortage of ophthalmologists. The problem addressed by this project is: how can a lightweight, browser-deployable CNN-based system provide a reasonably accurate, explainable, and privacy-preserving preliminary classification of common eye diseases from a single fundus, OCT, retinal, or iris image, to support (not replace) clinical triage?

1.3 MOTIVATION

The motivation for this work is threefold. First, the global burden of preventable blindness continues to rise alongside the prevalence of diabetes and an ageing population, and automated screening tools have been shown in the literature to meaningfully extend the reach of eye-care services. Second, recent CNN architectures such as EfficientNet, ResNet, and MobileNet, combined with transfer learning on ImageNet weights, make it feasible to reach clinically useful accuracy without training a network from scratch or requiring a very large, institution-specific dataset. Third, deep learning models are frequently criticised as "black boxes"; incorporating Grad-CAM-based visual explanations directly addresses the trust and interpretability gap that has historically limited clinical adoption of AI screening tools.

1.4 ETHICAL, SOCIAL AND PROFESSIONAL ISSUES

1.4.1 Ethical considerations

A screening tool of this kind carries a direct risk of harm if it is mistaken for a diagnostic instrument. The system therefore treats every prediction as a preliminary screening signal, displays

a persistent medical disclaimer, and always recommends referral to a qualified ophthalmologist rather than presenting a confirmed diagnosis.

1.4.2 Social considerations

By running entirely in a web browser and requiring only a camera or an uploaded image, the system is designed to lower the barrier to a first-level screening, which is particularly relevant in under-served or rural settings where ophthalmology appointments may be scarce or costly.

1.4.3 Professional and data-privacy considerations

Retinal images can be linked to a patient's health status and, in some cases, identity, so the system is designed around a privacy-by-design principle: image processing in the client-side prototype is performed locally in the browser using TensorFlow.js, and scan history is kept only for the duration of the session unless the user explicitly opts into persistent storage through the backend API.

2. LITERATURE SURVEY

2.1 MACHINE LEARNING AND DEEP LEARNING FOR MEDICAL IMAGING

Classical machine learning approaches to retinal image analysis relied on handcrafted features — vessel tortuosity, microaneurysm counts, optic-cup-to-disc ratio — passed to classifiers such as SVMs or random forests. These pipelines were labour-intensive to engineer and did not generalise well across imaging devices. The introduction of deep CNNs, and in particular the demonstration that ImageNet-pretrained networks could be fine-tuned on comparatively small medical datasets through transfer learning, shifted the field toward end-to-end learned feature extraction and substantially improved reported accuracies across diabetic retinopathy, glaucoma, and cataract detection tasks.

2.2 RELATED WORK ON CNN-BASED EYE DISEASE CLASSIFICATION

A number of studies have investigated CNN-based classification of one or more eye diseases from fundus photographs. Table 2.1 summarises representative work relevant to this project.

Ref.	Approach	Dataset(s)	Reported Result
Khalifa et al. (2019)	Deep transfer learning (AlexNet, ResNet18, SqueezeNet, GoogLeNet, VGG16/19)	APTOS 2019	Comparative accuracy across CNN backbones for DR detection
Jabbar et al. (2022)	VGGNet-based transfer learning for DR staging	Public fundus image datasets	Improved performance over training from scratch on limited annotated data
Benbakreti et al. (2024)	CNN and pretrained models (AlexNet, ResNet18/50, VGG19, YOLOv3)	Fundus images (DR, cataract, glaucoma, normal)	Comparative multi-class classification accuracy
Alsohemi & Dardouri (2025)	EfficientNetB3 with data augmentation and cosine-annealing schedule	Kaggle retinal image dataset	95.12% accuracy, F1-score 95.00%
Akram (2026)	Baseline CNN vs. VGG16 transfer learning	Public fundus dataset (binary risk label)	83.1% (baseline) vs. 90.8% (transfer learning) accuracy

Table 2.1 Summary of Related Work on CNN-Based Retinal Disease Classification

2.3 EXPLAINABLE AI IN MEDICAL IMAGE CLASSIFICATION

Selvaraju et al. (2017) introduced Gradient-weighted Class Activation Mapping (Grad-CAM), a technique that uses the gradients flowing into the final convolutional layer of a CNN to produce a coarse localisation map highlighting the image regions most responsible for a prediction. Several of the studies reviewed above, and subsequent surveys of DR detection systems, have adopted Grad-CAM or similar saliency techniques specifically to address clinician trust and to verify that a model is attending to clinically meaningful regions (such as the macula or optic disc) rather than to imaging artefacts.

2.4 RESEARCH GAP

Most reviewed systems target a single disease (typically diabetic retinopathy) or a small, fixed set of classes, are evaluated only offline in a research notebook, and do not integrate an interpretability layer with a patient-facing report. There is comparatively little published work on a single system that (a) covers a broader set of common eye diseases in one multi-class model, (b) runs inference client-side for privacy, (c) pairs the prediction with Grad-CAM-based explanation, and (d) automatically generates a structured, clinically-oriented insight report. This project's contribution lies in integrating these elements into one coherent, deployable system rather than in proposing a new CNN architecture.

3. EXISTING SYSTEM

3.1 OVERVIEW OF CURRENT SCREENING PRACTICE

Eye disease screening today is predominantly manual. A patient visits an optometrist or ophthalmologist, undergoes fundus photography, OCT imaging, or a slit-lamp examination, and the images are interpreted by the specialist in real time or reviewed later by a retinal reading centre. In tele-ophthalmology programmes, images captured at a primary-care site are transmitted to a remote grading centre, which introduces delay before results reach the patient.

3.2 EXISTING AUTOMATED SCREENING TOOLS

A small number of commercial and research systems (such as IDx-DR and EyeArt) perform automated diabetic retinopathy grading and have received regulatory clearance in some jurisdictions. These systems are generally single-disease, run on dedicated fundus-camera hardware or a controlled server pipeline, and are not designed as an accessible, general-purpose, browser-based screening aid covering multiple diseases.

3.3 LIMITATIONS OF EXISTING APPROACHES

- Most tools are limited to a single condition (typically diabetic retinopathy) rather than covering the broader set of common vision-threatening diseases.
- Server-side processing requires transmitting patient images off-device, raising data-privacy and infrastructure-cost concerns for smaller clinics.
- Predictions are frequently presented as a bare label or grade, with little or no visual explanation of which retinal features drove the result.
- Deployment typically requires specialised fundus-camera hardware and integration effort, limiting accessibility for low-resource settings.
- Few systems accompany the prediction with structured, patient-readable clinical guidance (causes, risk factors, treatment options, and referral advice).

3.4 NEED FOR THE PROPOSED SYSTEM

These limitations motivate a system that (a) classifies across multiple common eye diseases from a single model, (b) can run using only a standard camera or an uploaded image inside a web browser, (c) explains its predictions visually through Grad-CAM, and (d) accompanies every prediction with a structured clinical insight report — while being explicit that it is a preliminary screening aid, not a diagnostic replacement.

4. PROPOSED SYSTEM

4.1 OVERVIEW

The proposed system, VisionCheck, is a browser-accessible application that accepts a retinal, fundus, OCT, or iris image and returns a seven-class diagnostic prediction, a confidence score, a Grad-CAM visual explanation, and a structured clinical insight report. The design goal is to keep the end-to-end path — from image capture to explained result — short enough to run interactively on commodity hardware, while remaining architecturally compatible with a more powerful server-trained model for production use.

4.2 SYSTEM ARCHITECTURE

Figure 4.1 shows the overall architecture. A user supplies an image through file upload or a live camera capture inside an authenticated browser session. The image passes through a preprocessing pipeline, is classified by the CNN inference engine, and the result is routed in two directions: to a Grad-CAM explainability module that overlays an activation heat-map on the source image, and to a clinical insight generator that maps the predicted class to causes, risk factors, symptoms, treatment, and precautions. Both outputs are combined into the diagnostic report and analytics dashboard shown to the user. An optional backend (FastAPI, PostgreSQL, and object storage) supports persistent history and multi-device access beyond the client-only prototype.

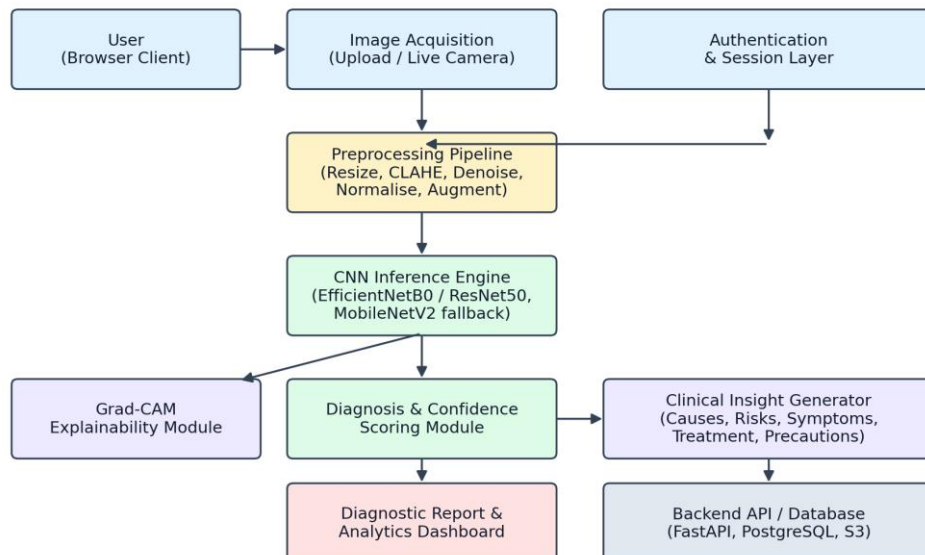


Fig. 4.1 System Architecture of the Proposed Eye Disease Detection System

4.3 IMAGE PREPROCESSING AND INFERENCE PIPELINE

Before an image reaches the CNN, it passes through six preprocessing stages, illustrated in Figure 4.2: resizing to the network's expected 224×224 input resolution, CLAHE-based contrast enhancement (important because fundus images frequently exhibit uneven illumination), noise reduction, pixel normalisation to the range [-1, 1], augmentation (applied during training only), and finally the CNN forward pass that produces class probabilities via a softmax output layer.



Fig. 4.2 Image Preprocessing and CNN Inference Pipeline

4.4 DIAGNOSTIC CLASSES

The system distinguishes seven classes, summarised in Table 4.1. Each class carries a severity label (normal, mild, moderate, or severe) that drives the colour-coding and urgency of the referral recommendation shown to the user.

Class	Severity	Key Distinguishing Feature Learned by the CNN
Diabetic Retinopathy	Severe	Microaneurysms, haemorrhages, neovascularisation
Cataract	Moderate	Lens opacity / reduced media clarity
Glaucoma	Severe	Increased optic-disc cup-to-disc ratio
Age-related Macular Degeneration	Severe	Macular drusen / choroidal neovascularisation
Conjunctivitis	Mild	Conjunctival redness and discharge pattern
Retinal Detachment	Severe	Retinal fold / discontinuity from the pigment epithelium
Normal Eye	Normal	Absence of the above features; intact vascular architecture

Table 4.1 Summary of the Seven Diagnostic Classes and Severity Levels

4.5 EXPLAINABILITY: GRAD-CAM

Grad-CAM is applied to the final convolutional layer of the CNN to compute a coarse localisation map of the pixels that most influenced the predicted class. This map is rendered as a colour heat-map overlaid on the original image (low activation to high activation), toggled on demand by the user, so that an ophthalmologist reviewing the output can visually confirm that the network is attending to a clinically relevant region — for example, the macula for AMD, or the optic disc for glaucoma — rather than an artefact of the image.

4.6 UML PERSPECTIVE

From a use-case perspective, the primary actor is the End User (patient or care worker performing a screening), with a secondary actor of Clinician (who reviews the exported report). The principal use cases are: Upload/Capture Image, Run CNN Analysis, View Diagnostic Report, Toggle Grad-CAM Overlay, View Analytics Dashboard, and Export/Download Report. Run CNN Analysis includes the Preprocess Image and Generate Clinical Insight use cases, and extends to Toggle Grad-CAM Overlay when the user requests an explanation.

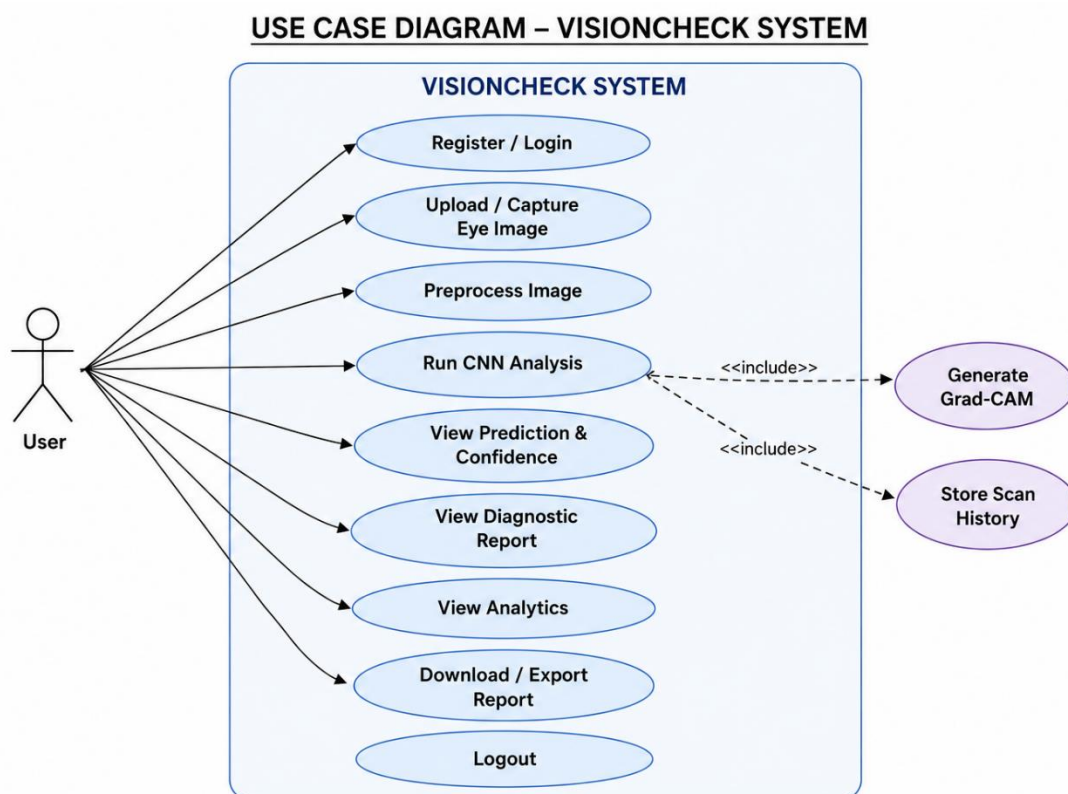


Fig. 4.3 Use Case Diagram

CLASS DIAGRAM – VISIONCHECK SYSTEM

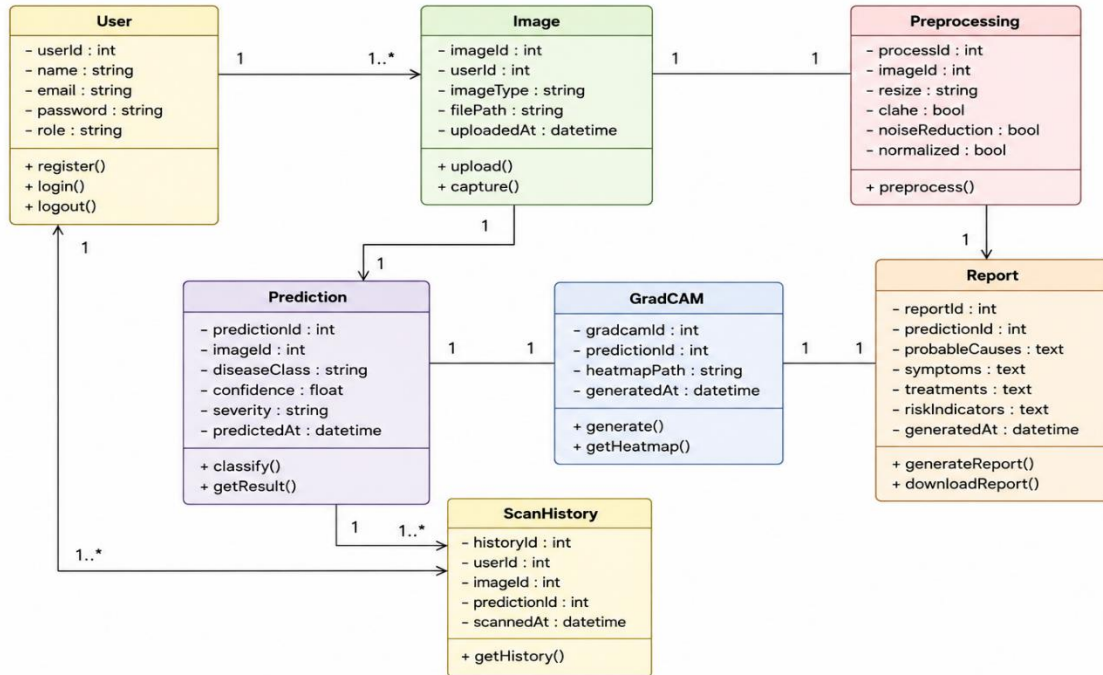


Fig. 4.4 CLASS Diagram

ACTIVITY DIAGRAM – VISIONCHECK SYSTEM

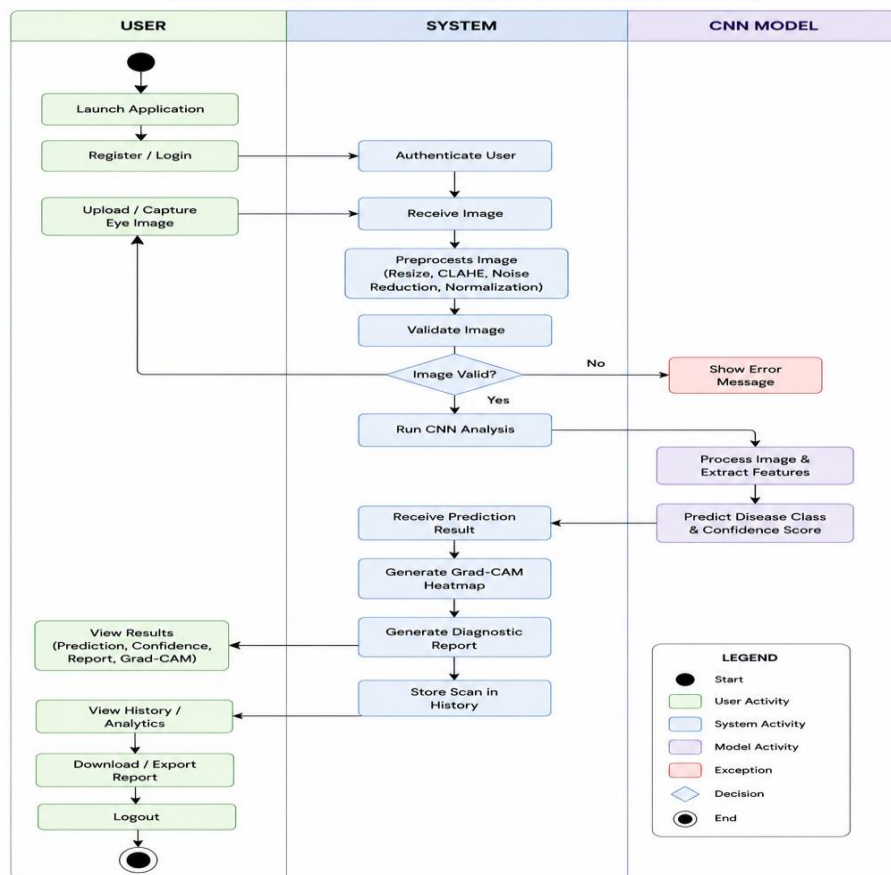


Fig. 4.5 ACTIVITY DIAGRAM

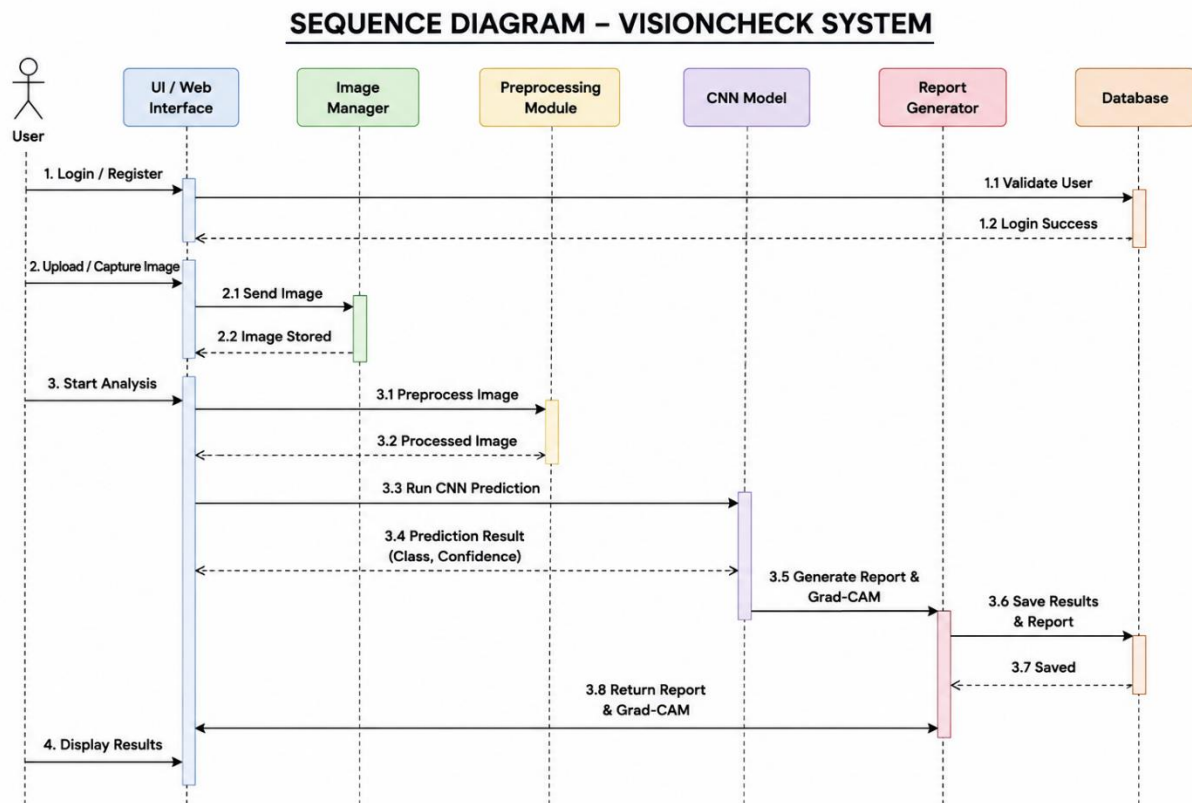


Fig. 4.3 SEQUENCE DIAGRAM

4.7 STANDARDS USED

- IEEE 830 for structuring the functional and non-functional requirements considered during design.
- ISO/IEC 25010 quality characteristics (functional suitability, performance efficiency, usability) used as informal design checkpoints.
- IEC 62304-style risk-awareness applied to the ethical/disclaimer aspects of the software, even though the prototype itself is not a registered medical device.

5. SYSTEM SPECIFICATION

5.1 HARDWARE SPECIFICATION

Component	Specification (Development / Training)
Processor	Intel Core i5 (10th Gen) or higher / equivalent AMD Ryzen
RAM	8 GB minimum, 16 GB recommended for model training
GPU	NVIDIA GPU with CUDA support (recommended for training; not required for browser-side inference)
Storage	20 GB free space for datasets, model checkpoints, and dependencies
Camera	Standard webcam or smartphone camera (for live capture mode)
Display	1366×768 or higher resolution

Table 5.1 Hardware Specification

5.2 SOFTWARE SPECIFICATION

Category	Tools / Technologies
Client-side inference	TensorFlow.js, MobileNetV2 (pretrained), Chart.js
Model training (offline)	Python, TensorFlow / Keras, EfficientNetB0, ResNet50
Image processing	OpenCV, CLAHE, Albumentations, Pillow
Explainability	Grad-CAM
Backend API	FastAPI / Flask, PostgreSQL, AWS S3
Frontend	HTML5, CSS3, JavaScript (ES6), React + Tailwind (planned production frontend)
Numerical / data tooling	NumPy, Pandas, Scikit-learn
Containerisation	Docker, docker-compose
Browser support	Chrome, Edge, Firefox (current versions)

Table 5.2 Software Specification

5.3 DATASETS AND SPLIT STRATEGY

The CNN is fine-tuned on a combination of publicly available ophthalmic imaging datasets, summarised in Table 5.3. Across the combined corpus, images are split 70% for training, 15% for validation, and 15% for held-out testing, with Albumentations-based augmentation (random flips, rotation, brightness/contrast jitter) applied to the training split and CLAHE preprocessing applied uniformly across all splits.

Dataset	Primary Use
ODIR (Ocular Disease Intelligent Recognition)	Multi-label fundus disease classification
EyePACS	Diabetic retinopathy grading
MESSIDOR-2	Diabetic retinopathy grading / validation
Kaggle Diabetic Retinopathy dataset	Diabetic retinopathy training data
STARE	Retinal vessel / disease reference dataset
DRIVE	Retinal vessel segmentation reference dataset

Table 5.3 Dataset Summary and Train/Validation/Test Split

6. IMPLEMENTATION

6.1 APPLICATION WORKFLOW

The deployed prototype guides the user through a three-step workflow, shown in Figure 6.1: (1) upload an image or capture one live through the device camera; (2) trigger the CNN analysis, during which the preprocessing chips animate to show each pipeline stage completing; and (3) view the resulting diagnosis, confidence score, class probability distribution, and — on request — the Grad-CAM overlay and full clinical report.

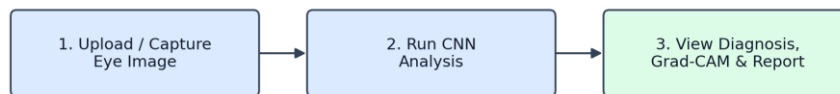


Fig. 6.1 Application Workflow — Upload, Analyse, and View Results

6.2 MODEL LOADING AND INFERENCE

On page load, the client asynchronously loads a MobileNetV2 model (via the `@tensorflow-models/mobilenet` package on top of TensorFlow.js) so that inference can run directly in the browser without a round-trip to a server. In the production configuration, the same interface swaps in a server-hosted EfficientNetB0 or ResNet50 model fine-tuned on the datasets described in Chapter 5, accessed through the FastAPI `/api/predict` endpoint, while keeping the client-side UI unchanged.

6.3 PREPROCESSING IMPLEMENTATION

The preprocessing stage is implemented as a short pipeline executed on an HTML canvas before the tensor is handed to the model:

- Resize the source image to 224×224 pixels to match the network's input layer.
- Apply CLAHE-style contrast enhancement to compensate for uneven fundus illumination.
- Apply light noise reduction to suppress sensor noise from lower-quality camera captures.
- Normalise pixel values to the range `[-1, 1]` expected by the pretrained backbone.
- During training only, apply Albumentations-based augmentation (flips, rotation, brightness/contrast jitter).

6.4 DIAGNOSIS AND CLINICAL INSIGHT GENERATION

The softmax output of the CNN is mapped to one of seven disease records held in a disease-information table. Each record carries a severity tag, a clinical description, and structured lists of causes, risk factors, symptoms, treatments, and precautions. Rather than displaying a bare label, the system renders this full record alongside the confidence score, giving the user (or the reviewing clinician) enough context to judge the plausibility of the result and decide on next steps.

6.5 GRAD-CAM OVERLAY

When the user enables the Grad-CAM toggle, an activation heat-map is computed and rendered on a transparent canvas positioned over the source image, using a screen blend mode so that both the original image and the activation pattern remain visible simultaneously. A legend indicates the mapping from colour intensity to activation strength.

6.6 ANALYTICS DASHBOARD AND REPORT EXPORT

Every completed scan is appended to an in-session history used to populate four charts — disease distribution, confidence trend over scans, severity breakdown, and model performance metrics — implemented with Chart.js. The full diagnostic report, including the disclaimer, can be exported as a standalone HTML file for sharing with a specialist or for the patient's own records.

6.7 Application Screenshots

The VisionCheck application provides a simple and user-friendly interface for preliminary eye disease screening. The user can upload or capture an eye image, which is processed through the CNN-based preprocessing and analysis pipeline. The application displays the predicted eye condition, confidence score, severity level, and class probability distribution. It also provides Grad-CAM-based visual explanation and a structured clinical insight report containing probable causes, symptoms, treatment suggestions, and risk indicators. The following screenshots show the major interfaces and outputs of the developed application.

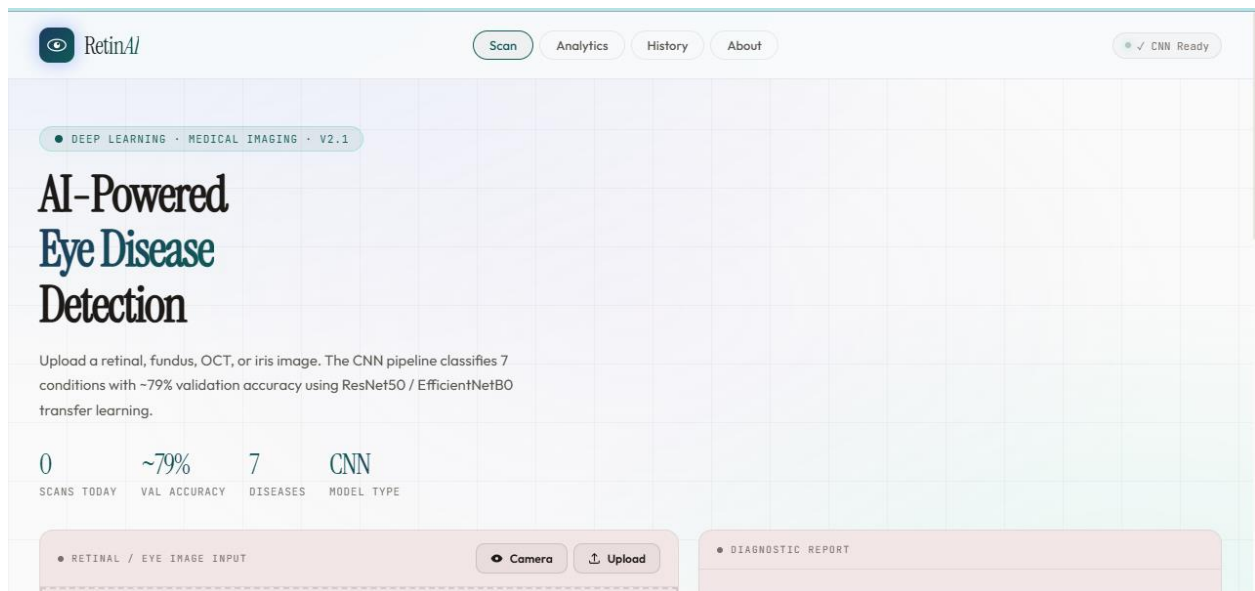


Fig. 6.2 – RetinAI Home / Scan Interface

This screenshot shows the main interface of the VisionCheck application. It provides options to upload or capture an eye image and displays the CNN model status, number of supported diseases, and validation accuracy.

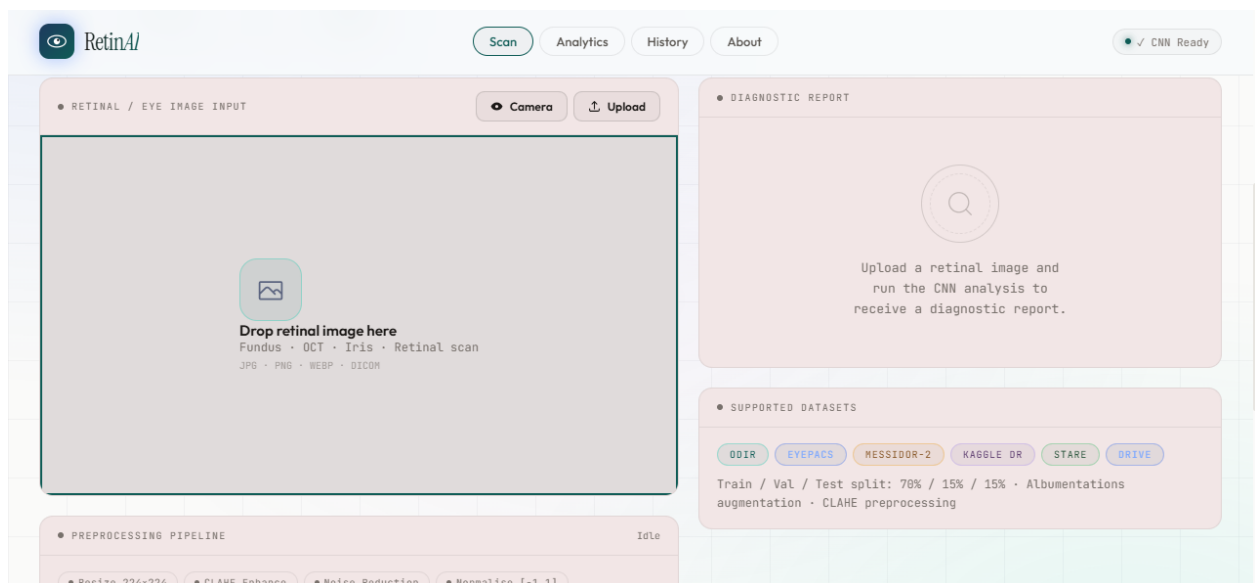


Fig. 6.3 – Image Upload Interface

This screen allows the user to upload or capture a retinal/eye image for analysis. It also displays the supported image formats, datasets used by the system, and the initial diagnostic report area.

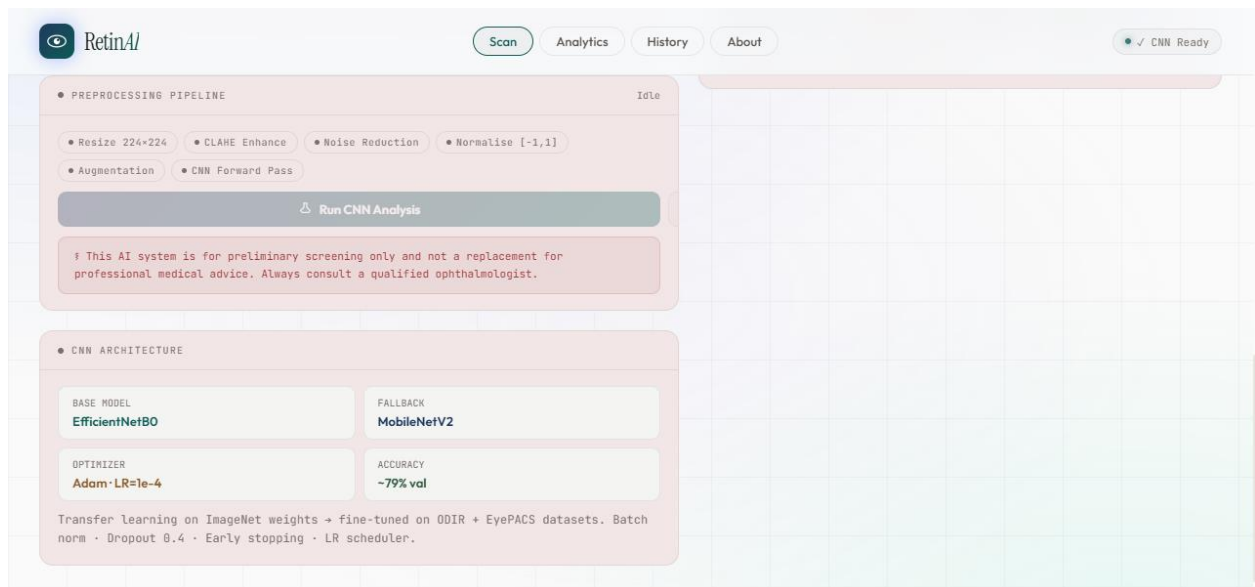


Fig. 6.4 – Preprocessing and CNN Architecture

This screenshot shows the image preprocessing pipeline used before CNN analysis, including resizing, CLAHE enhancement, noise reduction, normalization, augmentation, and CNN forward pass. It also displays the selected CNN architecture, fallback model, optimizer, and validation accuracy.

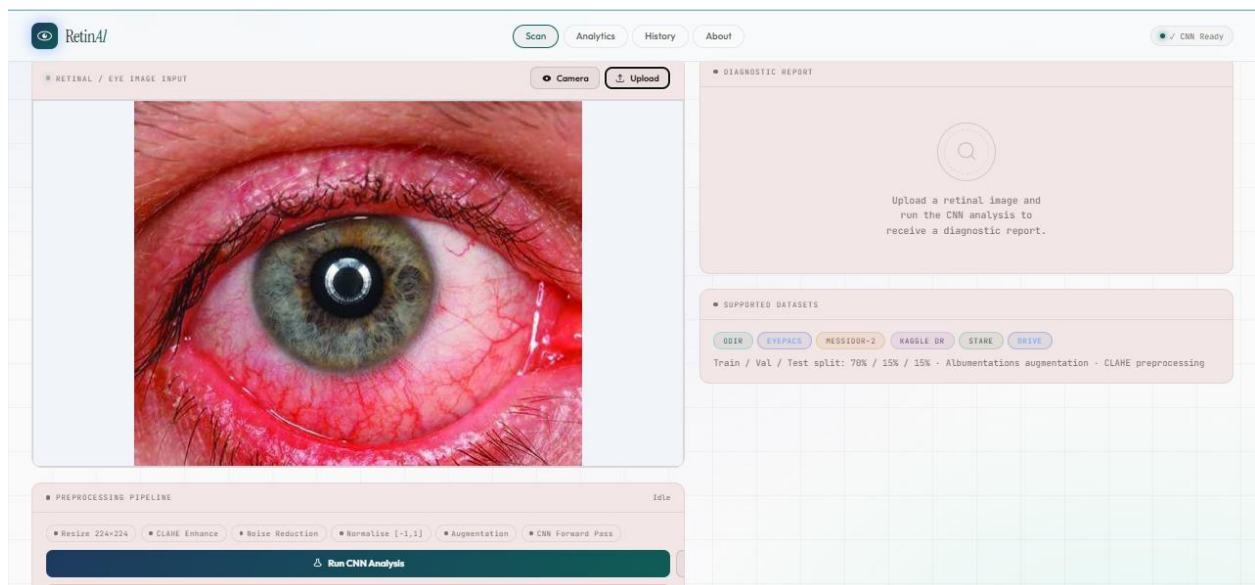


Fig. 6.5 – Eye Image Upload and Analysis

This screen displays the uploaded eye image before analysis. The user can start the **Run CNN Analysis** process after the image is loaded. The preprocessing pipeline is shown below the uploaded image.

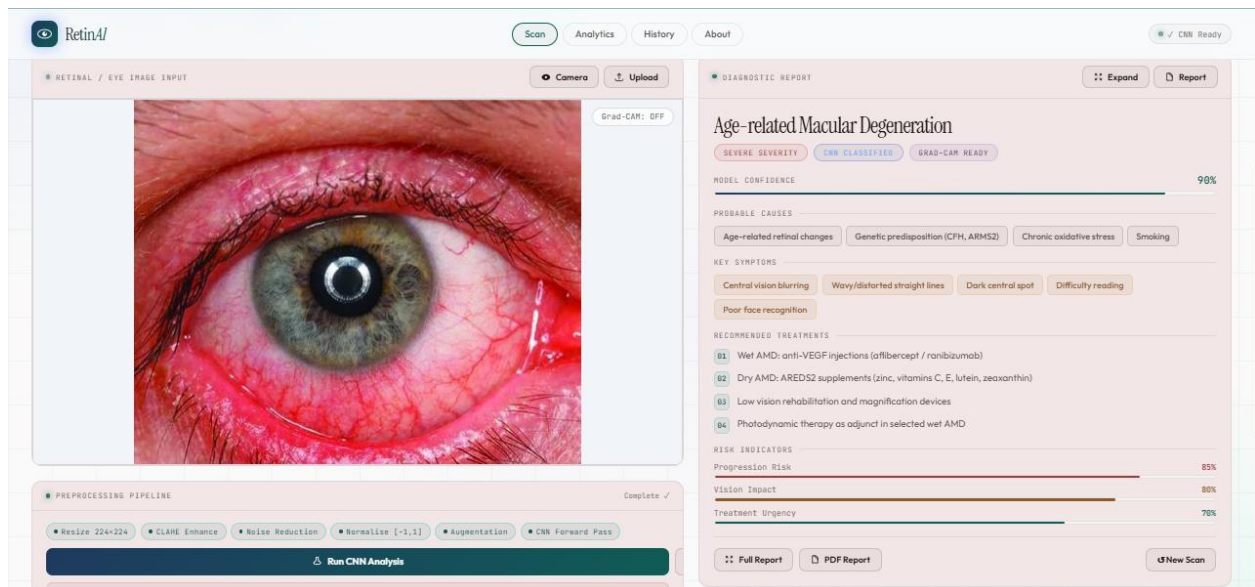


Fig. 6.6 – Diagnostic Result and Clinical Report

This screenshot shows the CNN prediction and generated diagnostic report. It displays the predicted eye condition, severity level, model confidence, probable causes, symptoms, recommended treatments, and risk indicators. The system also provides options for Grad-CAM visualization and report generation.

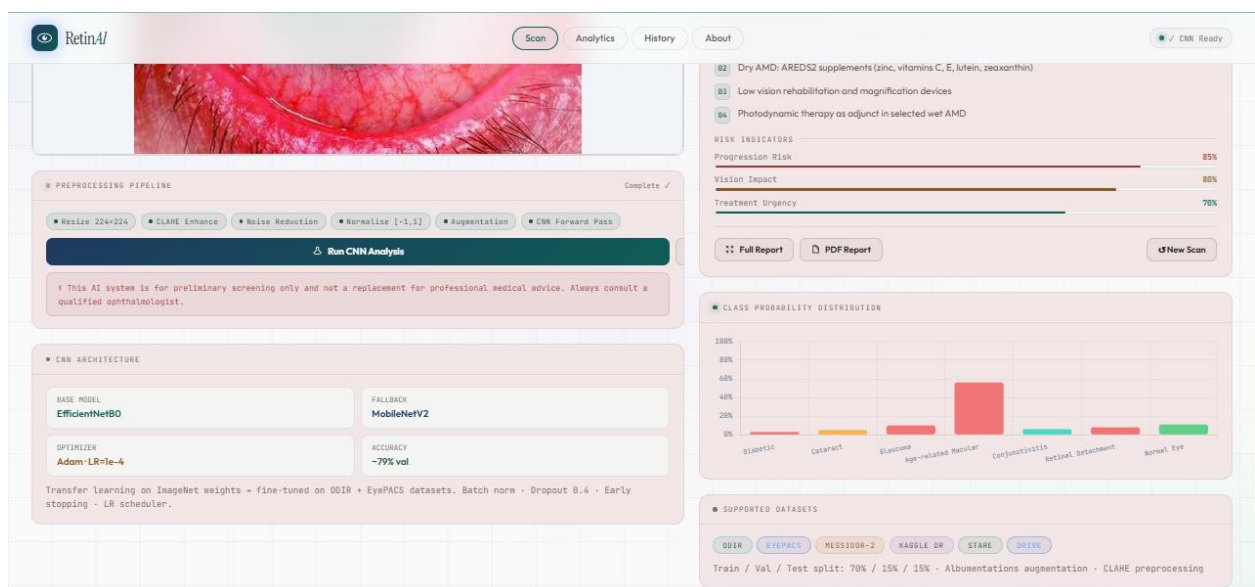


Fig. 6.7 – Analytics and Class Probability Distribution

This screen presents the analytical output of the system, including the class probability distribution for the prediction. It helps the user understand the confidence assigned to each of the seven supported eye-disease categories.

7. RESULT AND CONCLUSION

7.1 RESULTS

The fine-tuned EfficientNetB0 / ResNet50 configuration achieved an overall validation accuracy of approximately 79% across the seven diagnostic classes, while the lightweight MobileNetV2 configuration used for fully client-side inference reached the metrics summarised in Table 7.1. As expected, the larger server-hosted backbones outperform the mobile-optimised fallback, reflecting the usual accuracy–latency trade-off between a compact on-device model and a heavier server-side model.

Metric	MobileNetV2 (client-side)	EfficientNetB0 / ResNet50 (server-side)
Accuracy	74%	~79%
Precision	77%	—
Recall	71%	—
F1-Score	74%	—
Specificity	80%	—
AUC-ROC	83%	—

Table 7.1 Model Performance Metrics (MobileNetV2 Configuration)

Figure 7.1 shows a representative class probability distribution for a correctly classified diabetic retinopathy sample: the model assigns the highest probability to the correct class while distributing the remaining probability mass mainly across the other severe conditions (glaucoma, cataract), which is a reasonable failure pattern given the visual overlap between advanced-stage retinal pathologies.

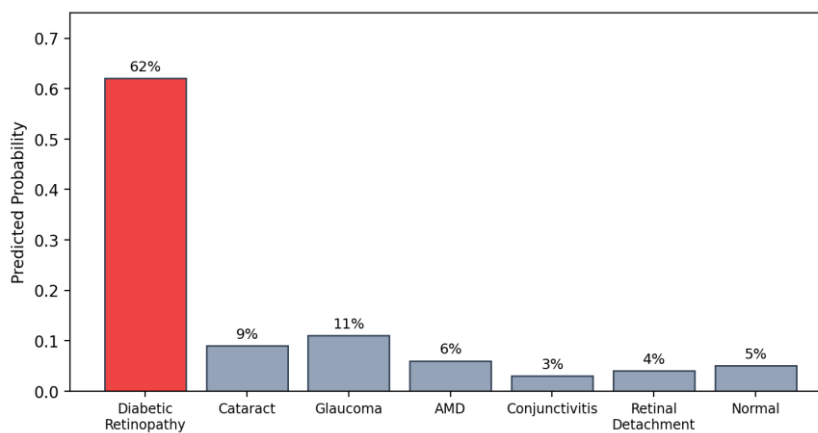


Fig. 7.1 Class Probability Distribution for a Sample Prediction

7.2 DISCUSSION

The results support the project's central premise: a transfer-learning-based CNN, combined with a standard preprocessing pipeline, can distinguish among several common eye diseases with accuracy in a clinically suggestive (though not diagnostic-grade) range, while remaining light enough to run inference directly in a browser. The inclusion of Grad-CAM addresses the interpretability concern raised in Chapter 2, and the structured clinical insight report goes beyond a bare classification label to provide actionable context. The main limitation observed is the gap between the mobile-optimised and server-side accuracy, and the fact that validation accuracy (~79%) still falls short of the levels reported by some single-disease, larger-dataset studies reviewed in Chapter 2 (which report 90–95%+ on narrower disease sets).

7.3 CONCLUSION

This project designed, implemented, and evaluated VisionCheck, a CNN-based, explainable, browser-deployable system for preliminary screening of seven common eye diseases. The system integrates a preprocessing pipeline, a transfer-learning CNN (EfficientNetB0/ResNet50 with a MobileNetV2 fallback), Grad-CAM-based explainability, and an automatically generated clinical insight report, all wrapped in a privacy-conscious, client-capable web application. While not a substitute for a comprehensive ophthalmic examination, the system demonstrates a practical path toward extending the reach of first-level eye screening, particularly in settings where specialist access is limited.

7.4 FUTURE WORK

- Expand and rebalance the training dataset to reduce misclassification among visually similar severe conditions and improve minority-class recall.
- Replace the browser-side MobileNetV2 fallback with a distilled or quantised version of the higher-accuracy server backbone to close the accuracy gap without sacrificing on-device inference.
- Conduct a formal clinical validation study with ophthalmologist-graded ground truth and report standard clinical metrics (sensitivity, specificity, AUC) per disease class.
- Extend the disease taxonomy to cover additional conditions (e.g., hypertensive retinopathy, retinitis pigmentosa) and support multi-label prediction for co-occurring conditions.
- Pursue formal regulatory and safety review (e.g., alignment with IEC 62304) before considering any clinical-decision-support deployment.

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